DATA SCIENCE AND VISUALIZATION

Project Planning and Dataset Selection

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team Number: Team 9

1. INTRODUCTION

The first week focused on team formation, dataset selection, defining the problem statement and objectives, and planning the project workflow. The team selected a Spotify tracks dataset containing information about songs, artists, albums, popularity, and audio features.

The team also learned about Pandas and NumPy for data analysis and identified the tools required for the project. These activities established the foundation for exploring the relationship between musical characteristics and song popularity in the upcoming stages.

2. PROJECT TITLE

Discovering the Factors Behind Song Popularity Through Data Science

3. PROBLEM STATEMENT

Music streaming platforms contain a large number of songs with different musical characteristics and popularity levels. Understanding the characteristics associated with song popularity can help reveal meaningful patterns in music data.

This project aims to analyze Spotify track data using data science and visualization techniques. The project will involve understanding the dataset, assessing its quality, handling missing values and inconsistencies, performing exploratory data analysis, and investigating relationships between audio features and popularity.

The findings will be communicated through suitable visualizations and, in the later stages, an interactive dashboard.

4. OBJECTIVES

The main objectives of the project are:

- To select and understand the Spotify dataset and its attributes.
- To learn Pandas and NumPy for data analysis.
- To identify data-quality issues and perform exploratory data analysis.
- To analyze relationships between musical features and popularity, create visualizations, and prepare data for advanced analysis and dashboard development.

5. TEAM FORMATION

The project team consists of four members who will collaboratively work on dataset preparation, analysis, visualization, documentation, and presentation.

S. No.	Team Member	Register Number
1	Prithesa R	25BAD083
2	Saravanan A	25BAD102
3	Tejas T	25BAD124
4	Viknesh T	25BAD128

The team will share responsibilities to ensure that each stage of the project is completed systematically. Specific individual task assignments can be documented once finalized by the team.

6. DATASET SELECTION

6.1 Dataset Description

The team selected the Spotify tracks dataset, containing 19,335 records and 25 columns covering song, artist, album, popularity, and audio characteristics.

6.2 Important Attributes

The dataset contains the following categories of information:

- **Track information:** Track ID, track name, track number, and album details.
- **Artist information:** Artist names associated with tracks.
- **Album information:** Album name, album type, release date, and number of tracks in an album.

- **Popularity information:** Popularity values associated with tracks.
- **Audio features:** Danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.
- **Musical attributes:** Key, mode, and time signature.

These attributes provide a combination of numerical, categorical, and textual data that can be examined using statistical and visualization techniques.

6.3 Reasons for Dataset Selection

The Spotify dataset was selected for the following reasons:

1. Represents real-world music data with details about songs, artists, and popularity.
2. Contains multiple attributes and audio features suitable for statistical and comparative analysis.
3. Provides opportunities to practise data cleaning, preprocessing, exploratory data analysis, correlation analysis, and visualization.
4. Supports advanced techniques such as feature engineering and Principal Component Analysis (PCA).

7. INTRODUCTION TO PANDAS

7.1 Definition

Pandas is an open-source Python library used for data manipulation and analysis. It provides two important data structures: Series and DataFrame.

A Series represents one-dimensional labelled data, whereas a DataFrame represents two-dimensional tabular data arranged in rows and columns.

7.2 Applications of Pandas

Pandas provides several useful operations for working with datasets:

1. Loading datasets from CSV and other file formats and displaying the first and last few records.
2. Examining the number of rows and columns, column names, and data types.
3. Identifying missing values and duplicate records and generating descriptive statistics.
4. Filtering, sorting, and transforming data for further analysis.

7.3 Application in Our Project

Pandas will be used to load and inspect the Spotify dataset. It will help the team understand the dataset structure, identify data-quality issues, and prepare the data for cleaning and exploratory analysis.

Example:

```
import pandas as pd
df = pd.read_csv("DSV_TEAM-9_RAW_DATASET.csv")
print(df.head())
print(df.shape)
```

The read_csv() function loads the dataset, head() displays the first five rows by default, shape returns the number of rows and columns.

8. INTRODUCTION TO NUMPY

8.1 Definition

NumPy, which stands for Numerical Python, is a Python library used for numerical computing. It provides the multidimensional array structure and mathematical functions required for efficient numerical operations.

8.2 Applications of NumPy

NumPy supports:

- Creating and manipulating numerical arrays for data processing.
- Performing mathematical and statistical calculations, including mean, median, and standard deviation.
- Carrying out element-wise operations and numerical transformations.
- Supporting data preprocessing, statistical analysis, and future PCA workflows.

8.3 Application in Our Project

NumPy will be used to perform numerical calculations on relevant Spotify audio features. It can help calculate statistical measures and prepare numerical data for further analysis.

Example:

```
import numpy as np
audio_features = np.array([0.75, 0.82, 0.68, 0.90, 0.73])
```

```
print("Average Audio Feature:", np.mean(audio_features))
print("Standard Deviation:", np.std(audio_features))
```

The `array()` function creates a numerical array, `mean()` calculates its average, and `std()` calculates its standard deviation.

9. PROPOSED PROJECT METHODOLOGY

The project will follow a systematic data science workflow to study Spotify song popularity and explore relationships between musical characteristics and popularity.

The planned workflow is:

1. **Raw Dataset Selection:** Select the Spotify tracks dataset for the project.
2. **Data Understanding:** Study the dataset structure, attributes, and meaning of the available information.
3. **Data Cleaning:** Plan the preparation of the dataset for reliable analysis.
4. **Exploratory Data Analysis:** Explore patterns and relationships among relevant variables.
5. **Feature Engineering:** Prepare and transform suitable attributes for analysis.
6. **Feature Selection:** Identify relevant features for investigating song popularity.
7. **Principal Component Analysis (PCA):** Explore dimensionality reduction where appropriate.
8. **Data Visualization:** Develop charts and visual representations of the findings.
9. **Insights and Data Storytelling:** Present the major findings in an understandable form.

12. TOOLS AND TECHNOLOGIES

The following tools and technologies are planned for the project:

- **Python:** Primary programming language for data analysis.
- **Pandas:** Data manipulation and handling tabular datasets.
- **NumPy:** Numerical calculations and mathematical operations.
- **Matplotlib:** Creating graphs and charts.
- **Seaborn:** Statistical data visualization.
- **Scikit-learn:** Supporting feature processing and PCA.
- **Jupyter Notebook:** Writing, executing, and documenting Python code.
- **GitHub:** Version control and team collaboration.
- **Streamlit or Plotly:** Potential tools for developing an interactive dashboard.

These tools will support the different stages of the project, from data preparation to visualization and presentation.

13. WEEK 1 ACTIVITIES AND MILESTONES

The main activities planned and undertaken during Week 1 were:

- Formed the project team, selected the Spotify tracks dataset, and finalized the project title.
- Defined the problem statement, project objectives, and overall project scope.
- Introduced Pandas and NumPy and identified the required software tools and libraries.
- Planned the project workflow, established GitHub for collaboration, and prepared the initial documentation.

These activities established the foundation for the subsequent stages of the project.

14. EXPECTED OUTCOMES

The expected outcomes of the project are:

- Understand the Spotify dataset and its attributes.
- Analyze the relationship between audio features and song popularity.
- Create visualizations to identify trends and important features.
- Develop an interactive dashboard to present the findings and insights.

15. CONCLUSION

Week 1 established the foundation for the Data Science and Visualization project titled *Discovering the Factors Behind Song Popularity Through Data Science*.

The team selected the Spotify tracks dataset, defined the problem statement and objectives, and planned the overall project methodology. The team also introduced Pandas and NumPy as essential Python libraries for data manipulation and numerical computation. GitHub was identified as the platform for maintaining project files and supporting collaboration.

These initial activities provide a clear direction for the project. In the following weeks, the team will proceed with the scheduled stages of dataset understanding, preparation, analysis, and visualization to investigate the factors associated with Spotify song popularity.

DATA SCIENCE AND VISUALIZATION-WEEK 2 REPORT

Know Your Data

Project Title: Discovering the Factors Behind Song Popularity Through Data Science

Team: 9

1. Introduction

Data understanding is an essential stage in the data science workflow. Before performing data cleaning, statistical analysis, or visualization, it is necessary to examine the structure, characteristics, and quality of the dataset.

During Week 2, the Spotify Songs dataset was explored using Python, Pandas, and NumPy. The analysis focused on inspecting the dataset, understanding its data types, calculating preliminary descriptive statistics, identifying missing values and duplicate records, and generating an automated First Impression Data Quality Report.

2. Aim

To load, inspect, and understand the structure and quality of the Spotify Songs dataset using Python, Pandas, and NumPy.

3. Objectives

1. Load the Spotify Songs dataset from a CSV file using Pandas.
2. Examine the dataset dimensions, first and last five records, column names, and data types.
3. Calculate preliminary descriptive statistics for numerical and categorical attributes.
4. Identify missing values, duplicate records, and potential data quality issues.
5. Generate an automated First Impression Data Quality Report and export the results as CSV files.

4. Tools and Technologies Used

- **Python:** Used for dataset exploration and analysis.
- **Pandas:** Used to load, inspect, and summarize the dataset.
- **NumPy:** Used for numerical operations.
- **Jupyter Notebook:** Used to execute the code and document the results.
- **CSV:** Used for storing the original dataset and exported reports.

5. Dataset Overview

The Spotify Songs dataset was obtained from Kaggle and contains information about songs, artists, albums, popularity, and audio characteristics.

The dataset includes attributes such as track name, artist information, album details, popularity, danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

Dataset details:

Attribute	Description
Total records	19,335
Total columns	46
File format	CSV
Data types	String, float, and integer
Memory usage	Approximately 74.65 MB

Each row represents a song record, while the columns contain information describing its metadata and audio features.

6. Dataset Loading and Inspection

The dataset was loaded into a Pandas DataFrame using `pd.read_csv()`. The `head()` and `tail()` functions were used to display the first five and last five records.

The `shape`, `columns`, `dtypes`, and `info()` functions were used to examine the dataset structure, column names, data types, non-null counts, and memory usage.

Observation: The dataset contains 19,335 rows and 46 columns. It includes both numerical audio features and textual metadata. Some attributes that represent numerical information are stored as strings and may require further validation.

7. Preliminary Descriptive Statistics

The `describe()` function was used to obtain an initial statistical summary of numerical and categorical attributes.

For numerical columns, the summary included count, mean, standard deviation, minimum, quartiles, and maximum. For categorical columns, it included the number of non-null entries, unique values, the most frequent value, and its frequency.

The numerical summary provided an initial understanding of the audio features. For example, the mean danceability was approximately **0.616**, mean energy was **0.584**, and mean tempo was approximately **119.79 BPM**.

Observation: The summaries provided an overview of the dataset's values and distributions. Detailed statistical analysis was left for a later stage of the project.

8. Missing Value Analysis

Missing values were identified using the `isnull()` and `sum()` functions. Missing percentages were calculated to evaluate the completeness of individual columns.

The analysis produced the following results:

Metric	Result
Total missing cells	7,707
Rows containing missing values	7,532
Columns containing missing values	39
Missing values in `preview_url`	7,524
Missing percentage in `preview_url`	38.91%

The `preview_url` column contained the largest number of missing values. Several audio feature and metadata columns also contained missing values, although some had only seven missing entries each.

Observation: Missing values were identified and documented without modifying the dataset. Appropriate handling decisions were reserved for the data cleaning stage.

9. Duplicate Record Analysis

The `duplicated()` function was used to identify exact duplicate rows. The Spotify song identifier column, `id`, was also checked separately for repeated identifiers.

The results were:

- **Exact duplicate rows:** 0
- **Repeated Spotify IDs:** 2

Observation: No exact duplicate rows were found. However, two repeated Spotify IDs were identified and require further investigation before deciding whether any records should be removed.

No records were deleted during Week 2.

10. Data Type Analysis

The data types of all columns were examined using `dtypes` and a data type summary table.

The dataset contained:

Data type	Number of columns
String (`str`)	32
Float (`float64`)	13
Integer (`int64`)	1
Total	**46**

The analysis also checked whether selected fields containing numerical information could be converted into numeric values.

Column	Invalid numerical values
`popularity`	1
`track_number`	1
`duration_ms.x`	4

Observation: Some numerical fields were stored as strings, and a small number of values could not be converted directly into numbers. These inconsistencies were documented for further examination. No data type corrections were made during Week 2.

11. Automated First Impression Data Quality Report

An automated First Impression Data Quality Report was generated using Pandas. It summarized the condition of all 46 columns and included:

- Column names and data types.
- Non-null counts.
- Missing value counts and percentages.
- Number of unique values.

A separate dataset summary was also generated to record the total rows, total columns, missing cells, rows containing missing values, columns containing missing values, exact duplicate rows, and memory usage.

The following CSV reports were exported successfully:

1. `Week2_Data_Quality_Report.csv`
2. `Week2_Dataset_Summary.csv`

Observation: The reports provide a structured overview of the dataset's completeness and quality. They will serve as references for planning the next stage of preprocessing.

12. Team Member Participation

The following is a proposed allocation of responsibilities based on the activities performed during Week 2.

S. No.	Team Member	Roll Number	Contribution
1	Prithesa R	25BAD083	Dataset loading, initial inspection, and understanding the dataset structure.
2	Saravanan A	25BAD102	Preliminary descriptive statistics and data type analysis.
3	Tejas T	25BAD124	Missing value analysis and duplicate record verification.
4	Viknesh T	25BAD128	Automated data quality report generation and report documentation.

13. Challenges and Considerations

1. The dataset contains missing values across multiple columns, particularly `preview_url`.
2. Some numerical attributes are stored as strings and require validation.
3. Repeated Spotify IDs need investigation even though no exact duplicate rows were found.
4. Metadata and URL fields require consideration when preparing the dataset for further analysis.

These findings were documented without changing the original dataset.

14. Conclusion

During Week 2, the Spotify Songs dataset was successfully loaded and examined using Python, Pandas, and NumPy. Its structure, data types, preliminary descriptive statistics, missing values, and duplicate records were analyzed.

The investigation identified 7,707 missing cells, no exact duplicate rows, two repeated Spotify IDs, and several values that require numerical validation. An automated First Impression Data Quality Report and a dataset summary were generated and exported successfully.

The original dataset was preserved without modification. The findings establish a foundation for the data cleaning and preprocessing activities in Week 3.